

PART A QUESTION-1

Date: _____

NAME: Bilal Yaway

STUDENT ID: 261K-2523

① Start

② Input N

③ Set i to N

④ While $i \geq 1$ repeat

 print i

 set i to $i-1$

⑤ end while

⑥ Stop

QUESTION-2

① Start

② Input year

③ If year is divisible by 400 then
 print leap year else if year is ~~no~~
 divisible by 100 print not leap year
 else if year is divisible by
 4 print leap year.

④ else print not leap year

⑤ Stop.

PART - A , QUESTION - 3

Date: _____

NAME: Bilal Yawar

STUDENT ID: 2612-2523

- ① Start
- ② Input N
- ③ set i to 0
- ④ while $N > 0$, repeat:
 - set $N = N/10$
 - set i to $i + 1$
- ⑤ if $i == 1$ print "Single Digit Number"
- ⑥ else if $i == 2$ print "Double Digit Number"
- ⑦ else print "Tripple Digit or more number"
- ⑧ Stop.

QUESTION - 4

- ① Start
- ② Set i to 1
- ③ while $i \leq 10$ repeat
 - if $i \% 3 \neq 0$, then
 - set $j = 1$
 - while $j \leq 10$, repeat
 - print $i \times j$
 - set $j = j + 1$
 - se end while
 - set $i = i + 1$
- ④ end while
- ⑤ stop.

PART A , QUESTION-5

Date: _____

NAME: Bilal Yawar

STUDENT ID: 26K-2523

- ① Start
- ② Input N
- ③ While $N \geq 10$, repeat
 - set sum to 0
 - while $N \geq 10$, repeat
 - set $i = N \% 10$
 - set $sum = sum + i$
 - set $N = N / 10$
 - print sum
 - set $N = sum$
- ④ print the final single digit value.
- ⑤ Stop.

NAME: Bilal Yawar

STUDENT ID: 26K-2523

- ① Begin
- ② Input season, room, nights
- ③ If season = peak, then
 - If room = standard, set rate to 5000
 - else if room = deluxe, set rate to 8000
 - else if room = suite, set rate to 12000
- ④ Else if season = off-peak, then
 - If room = standard, set rate to 3000
 - else if room = deluxe, set rate to 500
 - else if room = suite, set rate to 8000
- ⑤ Calculate cost = rate $\times$ nights
- ⑥ If nights > 7 then
 - calculate discount = cost $\times 0.15$
 - calculate total = cost - discount
- ⑦ otherwise, set total = cost
- ⑧ Display total
- ⑨ End.

PART. B

ELEVATOR SIMULATION

Date: _____

NAME: Bilal Yawar

STUDENT ID: 26K-2523

- ① Start
- ② set floor to 0
- ③ Input N
- ④ set i to 1
- ⑤ while $i \leq N$, repeat
Input floor requested
If floor requested > floor print Moving Up
Else if floor requested < floor print Moving down
Else, print Doors Opening
set floor to floor requested
set i to $i + 1$
- ⑥ End while
- ⑦ Stop.

PART-B

CLASS RESULT PROCESSING

Date: _____

NAME: Bilal Yawar

STUDENT ID: 2612-2523

- ① Start
- ② Input N
- ③ Set i to 1
- ④ while $i \leq N$ repeat
 set sum to 0
 set fail to 0
 set numsubject = 1
- ⑤ while numsubject ≤ 5 , repeat
 input marks
 set sum to sum + marks
 if marks < 33 , set fail to 1
 set numsubject to numsubject + 1
- ⑥ end while
- ⑦ calculate $avg = sum / 5$
- ⑧ If fail = 1 print "fail-Subject-Deficiency"
- ⑨ else if $avg \geq 80$, print "Distinction"
- ⑩ else if $avg \geq 60$, print "Pass"
- ⑪ else, display "fail".
- ⑫ set i to $i + 1$
- ⑬ end while
- ⑭ Stop.

PART B

ONLINE SHOPPING BILL CALCULATOR Date: _____

NAME: Bilal Yawar

Student ID: 26K-2523

① Start

② Input quantity, price, discount

③ If quantity ≤ 0 , print Invalid Quantity

④ else if price ≤ 0 , print Invalid Price

⑤ else if discount < 0 OR discount > 100 , print Invalid

Discount %.

⑥ else if tax < 0 OR tax > 100 print Invalid

Tax %.

⑦ else :

- calculate subtotal = quantity $\times$ price

- calculate discounted amount =

subtotal - (subtotal $\times$ discount) / 100

- calculate bill = discounted amount

+ (discounted amount $\times$ tax) / 100

- print subtotal.

- print discounted amount.

- display bill.

⑧ Stop.